

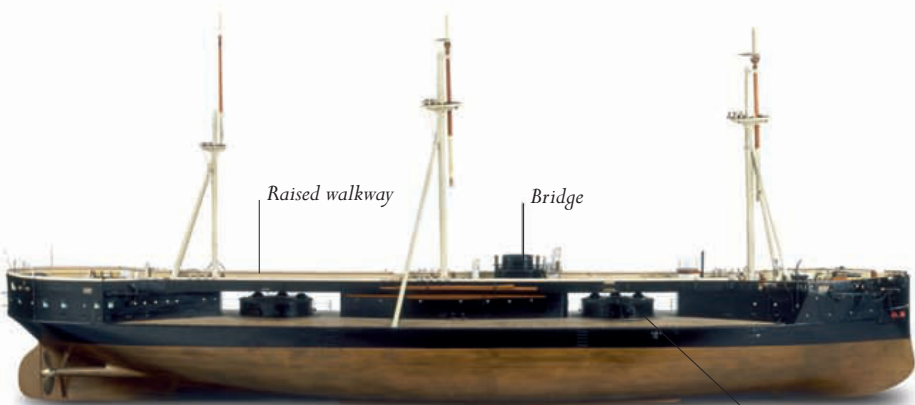
BATTLESHIPS

The second half of the 19th century was a time of considerable technological innovation at sea. Already the world's naval forces had begun to embrace steam propulsion, but now the very essence of the warship was to undergo a complete transformation, as wood and wind gave way to steel and steam. New types of weapons, as well as new ways of handling them, were developed and introduced. A dramatic transition occurred in a period of just 40 years, and in a world known for its innate conservatism and reluctance to accept change, this was no mean achievement.



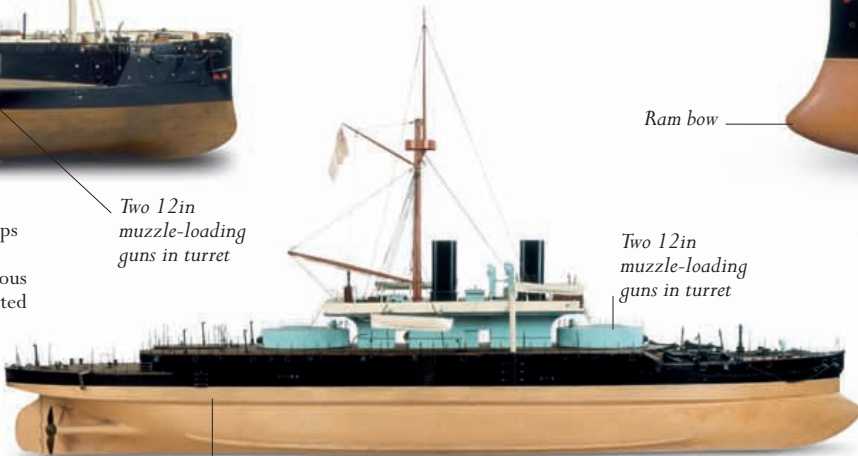
▲ HMS WARRIOR	
Commissioned	1861 Origin UK
Displacement	9,140 tons
Length	128m (420ft)
Top speed	14.1 knots

The *Warrior* and its sister-ship *HMS Black Prince* were the first ocean-going “ironclads”, with 11.5cm (4½in) of armour on 45.7cm (18in) wooden hulls, screw propellers as well as a full sailing rig, and breech-loading guns.



▲ HMS CAPTAIN	
Commissioned	1870 Origin UK
Displacement	7,770 tons
Length	97.5m (320ft)
Top speed	15.25 knots

An experimental ship produced to the design of Captain Cowper Phipps Coles, a pioneer of the gun turret, *HMS Captain* proved to be a disastrous mistake. Its low freeboard contributed to its loss during a severe storm in September 1870.



▲ HMS DEVASTATION	
Commissioned	1873 Origin UK
Displacement	9,330 tons
Length	93.6m (307ft)
Top speed	13.5 knots

Devastation and its sister-ship *Thunderer* were the first mastless sea-going battleships, with their armament contained in turrets fore and aft of the superstructure. This feature set the pattern for future development.



▲ HMS RUPERT	
Commissioned	1874 Origin UK
Displacement	5,440 tons
Length	80.5m (264ft)
Top speed	14 knots

Following the success of ramming tactics at the Battle of Lissa in 1866, many navies built ships specifically designed to sink others by ramming them. The British Royal Navy commissioned four, including *Rupert*, but none ever saw combat.

